Bahir Dar University

College of Education

Course: **Fundamentals of Electricity and Electronics Devices**

Lab report for Computer Science 2nd Year Students

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#Lab 1

Introduction

Resistors are essential components in electronic circuits, and there are two ways(methods) to know the value of the of the resistors :

1. Circulating the color code: using the color on the resistor and a table that give the value of the color we can calculate the resistance and tolerance.
2. Direct measurement: using and other device like a digital multimeter we can measure the value of the resister.

Objective

The aim of this experiment is to know how we can measures resistance using a device of using the color code on the resister it self and compare the theoreticanl value calculated from the color bands with the device measured value. This helps us to understand measurement accuracy, tolerance, and the reliability of the resistors.

Experimental procedure

A. Calculating resistance using the color band(code)

Count how much color band there are on the resister:

- 4 band resistors: this resistors have 4 colors the first two bands are digits(significant figures), the third one is the multiplier and the fourth one is the tolerance.
- 5 band resistors: this resistors have 5 colors the first three bands are digits(significant figures), the fourth one is the multiplier and the fifth one is the tolerance.
- 6 band resistors: this resistors have 5 colors the first three bands are digits(significant figures), the fourth one is the multiplier, the fifth one is the tolerance and the sixth or last one is the show the temperature coefficient in part per million per degree Celsius (ppm/^{0}C).

For 4 and 5 band resistors we should make the gold or silver the last color and begin on the opposite side and for 6 band resistors we should make the thick band the last and start at the opposite side.

Based on the band on the resistors convert the color to their values and calculate the resistance.

B. Measure resistance using a Multimeter

We can measure the resistance with a multimeter; we can measure it through any direction:

- First, we turn the multimeter on and switch its measuring value to resistor rang so that it can tell us the resistance value.
- Clean the resistors lead so that we can have a good connection.
- Next connect the multimeter probes to the resistor lead, don't touch both probes with your hands of you will get a false reading.
- Record the reading of the device.

Finally compare the calculated result to the measured result.

Experimental data / observation ✞ 4

band resistors:

Brown – Red – Yellow – Gold

We referred the values of the color form the table:

Brown = 1

Red = 2

Yellow = 10k

Gold = ± 5%

Calculation

R = 12 * 10k = 120k Ω ± 5%

Tolerance range

R_{min} = 120k Ω – (0.05*120k Ω) = 114000 Ω

R_{max} = 120k Ω + (0.05*120k Ω) = 126000 Ω

✞ So, the rang is 114000 Ω - 126000 Ω

The measured value for this resistor is 118.3k Ω so it is found within the tolerance rang.

✞ 4 band resistors:

Brown – Green – Green – Gold

We referred the values of the color form the table:

Brown = 1

Green = 5

Green = 100k

Gold = ± 5%

Calculation

R = 15 * 100k = 1500k Ω ± 5%

Tolerance range

R_{min} = 1500k Ω – (0.05*1500k Ω) = 1425000 Ω

R_{max} = 1500k Ω + (0.05*1500k Ω) = 1575000 Ω

✞ So, the rang is 1425000 Ω - 1575000 Ω

The measured value for this resistor is 1.550M Ω (1550000 Ω) so it is found within the tolerance rang.

✞ 5 band resistors:

Orange – Orange – Black – Orange – Brown

We referred the values of the color form the table:

Orange = 3

Orange = 3

Black = 0

Orange = 1k

Brown = ± 1%

Calculation

R = 330 * 1k = 330k Ω ± 1%

Tolerance range

R_{min} = 330k Ω – (0.01*330k Ω) = 326700 Ω

R_{max} = 330k Ω + (0.01*330k Ω) = 333300 Ω

✞ So, the rang is 326700 Ω - 333300 Ω

The measured value for this resistor is 330.2k Ω so it is found within the tolerance rang.

✞ 5 band resistors:

Brown – Black – Black – Yellow – Brown

We referred the values of the color form the table:

Brown = 1

Black = 0

Black = 0

Yellow = 10k

Brown = ± 1%

Calculation

R = 100 * 10k = 1M Ω ± 1%

Tolerance range

R_{min} = 1000k Ω – (0.01*1000k Ω) = 990000 Ω

R_{max} = 1000k Ω + (0.01*1000k Ω) = 1010000 Ω

✞ So, the rang is 990000 Ω - 1010000 Ω

The measured value for this resistor is 0.999M Ω (999000 Ω) so it is found within the tolerance rang.

✞ 6 band resistors:

Red – Red – Brown – Black – Brown – Red

We referred the values of the color form the table:

Red = 2

Red = 2

Brown = 1

Black = 1

Brown = ± 1%

Red = 50ppm

Calculation

R = 221 * 1 = 221 Ω ± 1%

Tolerance range

R_{min} = 221 Ω – (0.01*221 Ω) = 218.79 Ω

R_{max} = 221 Ω + (0.01*221 Ω) = 223.21 Ω

✞ So, the rang is 218.79 Ω - 223.21 Ω

The measured value for this resistor is 220.8 Ω so it is found within the tolerance rang.

Discussion and analysis

When ever we try to do calculate a 4 and 5 band resistor we should make the Gold color or the Silver the last and begin on the opposite side, and if the band is 6 we should make the thick band the last and start on the opposite side.

The calculation of the color code gives us the tolerance rang of the resistor and our calculation alines with our measures value.

Conclusion

The lab session successfully demonstrated two methods of determining resistor values:

- Using multimeter and theoretical (calculating the value of the resistor using the band

on the resister)

The experiment confirms that resistor tolerance plays a major role in determining how close actual values match theoretical value

#Lab 2

RESISTORS IN SERIES

THEORY: SERIES CIRCUITS

A series circuit is a type of electrical circuit where all components are connected end to end along a single path for a current flow .There is only one path for the current ,which makes the same current flow through every resistor.

->current in series(I)

Electrons have only one path to travel ,so no matter how many resistors are connected ,the same number of electrons pass through each resistor per second.

$I_{total}=I_1=I_2=...$

->Voltage in series (V)

The total voltage of the supply is divided among the resistors in proportions to their resistances.

Ohm's law: V=I.R

$V_{total}=V_1+V_2+V_3+...$

➔ Equivalent resistance

Is the sum of all individual resistances:

$R_T=R_1+R_2+R_3..$

OBJECTIVE

1.to study series circuit connection of the resistors in an electrical circuit.

2.To determine the total (equivalent) resistance of a series circuit.

3.verify that the total resistance equals the sum of individual resistances.

MATERIALS REQUIRED

*Different value resistors

*Circuit construction board

*Digital multimeter

*Connecting wires

PROCEDURE

a.measure each resistor values using digital multimeter.

b.Construct the series circuit on construction board.

c.calculate the circuit's total resistance(R_T) and record in a table .

d.measure the circuit's resistance and record in same table.

e.calculate the percentage difference between calculated and measured values.(compare experimental and theoretical results

f.remove components of circuit from board.

OBSERVATIONS AND CALCULATIONS

R 1 =0.102M ,R 2=0.471M ,R 3 =1.497M

SERIES	Circuit junction	Rcalculated	Rmeasured	%diff
	R_1-R_2	0.573M	0.572M	1.7*10^-6
	R_2-R_3	1.968M	1.968M	0
	R_1–R_3	1.599M	1.597M	1.2*10^-3
	RT	2.070M	2.070M	0

%DIFF=(R calculated-Rmeasured)/R calculated

CONCLUSION

-Experment shows that in series circuit ,the total resistance is sum of individual resistance.

-adding resistors in series increase total resistance.

-any break in series path stops the current entirely

REVIEW QUESTIONS

What do understand from the above experiment?

In series connection ,the total resistance is equal to sum of individuals.

2.if you want to get the smallest resultant resistance from two or more resistances,which type of connection you choose ?

To smallest resultant,aparallel connection should be used.

3.if you want to get the largest resultant resistance from 2 or more resistance ,which type of connection do you choose?

To the largest resultant resistance ,a series connection should be used.

4.describe the advantage and disadvantage of series connection .

<u>Advantage of series connection</u>

-simple to connect

-same current flows through all components

-useful when current control is required

<u>Disadvantage</u>

-if one component fails ,the entire circuit stops working

-not suitable for domestic wiring

#lab 3 and 4

Verification of Ohm's Law

Introduction / Theory

Electric current is defined as the rate of flow of electric charge in a circuit and is denoted by . The potential difference between two points in a circuit is known as voltage and is denoted by . Resistance, denoted by , is the opposition offered by a conductor to the flow of electric current.

According to Ohm's Law, for a metallic conductor at constant temperature, the current flowing through it is directly proportional to the voltage applied across it. Mathematically, Ohm's Law is expressed as:

I= V/R
or
V= IR

where is the constant of proportionality and represents the resistance of the conductor. Ohm's Law is widely used in electrical and electronic circuit analysis, especially for DC circuits where current flow is steady. In AC circuits, the concept of impedance is used instead of resistance, but the fundamental relationship between voltage and current remains significant.

Objectives

The objectives of this experiment are:

To study Ohm's Law experimentally.

To verify that the current through a DC circuit is directly proportional to the applied voltage.

To determine the resistance of a given resistor using measured values of voltage and current.

To study the relationship between voltage, current, and resistance in both DC and AC circuits.

Materials and Equipment

The following equipment were used in this experiment:

1. Digital multimeter (used as voltmeter, ammeter, and ohmmeter)

2. DC power supply

3. Fixed resistor

4. Circuit construction board

5. Connecting wires

Procedure

1. The resistance of the given resistor was first measured using a digital multimeter (ohmmeter).

2. The DC power supply was adjusted to deliver approximately 1 V.

3. The DC circuit was constructed as shown in Figure (a), with the ammeter connected in series and the voltmeter connected across the resistor.

4. The voltage across the resistor () was measured and recorded.

5. The corresponding current flowing through the circuit was measured using the ammeter.

6. Steps 2–5 were repeated by increasing the source voltage from 1 V up to 8 V in steps of 1 V.

7. The calculated current was obtained using Ohm’s Law and compared with the measured current values.

8. The same procedure was repeated for the AC circuit shown in Figure (b) using an AC supply.

5. Observations and Data (DC Circuit)

R= 219.6 ohm

Measured resistance of the resistor:

Source Voltage (V)	Voltage Across Resistor VR(v)	Calculated Current I=V/R (A)	Measured Current (A)
1	0.998	0.004555	0.046
2	1.996	0.00909	0.090
3	2.993	0.01363	0.130
4	3.992	0.01818	0.018
5	4.989	0.02272	0.023
6	5.988	0.02728	0.027
7	6.980	0.03178	0.032
8	7.980	0.03636	0.036

Sample Calculation

For :VR=0.998, V:

I=V/R=0.998÷219.6=0.00455,A

Similarly, the current for other voltage values was calculated using the same formula.

Result

From the experimental observations, it was found that the measured current values were in close agreement with the calculated current values obtained using Ohm's Law. As the applied voltage increased, the current increased proportionally, indicating a linear relationship between

voltage and current. The ratio remained approximately constant and equal to the resistance of the resistor. Thus, Ohm's Law is verified for the given resistor.

Conclusion

The experiment successfully verified Ohm's Law. The relationship between voltage and current was found to be linear, and the resistance of the given resistor remained nearly constant at . Minor deviations may be attributed to instrument accuracy, contact resistance, and temperature variations.

Precautions

All connections were made tight and secure.

The power supply voltage was increased gradually.

The resistor was not allowed to overheat.

Proper ranges of the multimeter were selected before taking measurements.

Sources of Error

Internal resistance of measuring instruments.

Loose connections in the circuit.

Fluctuations in power supply voltage.

Temperature changes affecting resistance.

Laboratory #4: Network Analysis Using Kirchhoff's Theorem

Theory

Kirchhoff's laws are fundamental tools used in electrical network analysis.
Kirchhoff's Voltage Law (KVL): The algebraic sum of all voltages around any closed loop in a circuit is zero.
Kirchhoff's Current Law (KCL): The algebraic sum of currents at a junction is zero.

Objectives

- To study Kirchhoff's Voltage and Current Laws in practical circuits.
- To analyze current distribution in an electrical network.
- To verify theoretical calculations with measured values.

Materials

- 5 resistors (R_1 = 356.6 Ω, R_2 = 177.1 Ω, R_3 = 177.4 Ω, R_4 = 218.8 Ω, R_5 = 261.2 Ω)
- Circuit construction board
- Digital multimeter (yellow device) for voltage measurements
- Connecting wires
- Power supplies: V_1 ≈ 9.9 V, V_2 ≈ 23.97V

Procedure

- Measure resistances of all 5 selected resistors using a digital multimeter.
- Construct the circuit on the construction board as shown in the lab diagram.
- Connect the 9.9 V source and measure voltage across each resistor (VR_1, VR_3, VR_4) using the digital multimeter.
- Calculate the sum of voltages in Loop 1 to verify KVL.
- Connect the 23.97 V source and measure voltage across each resistor (VR_2, VR_3, VR_5) using the digital multimeter.
- Calculate the sum of voltages in Loop 2 to verify KVL.
- Reverse the polarity of the 10 V source if required and repeat measurements.

Observations

- Measured Resistances:

R_1 = 356.6 Ω, R_2 = 177.1 Ω, R_3 = 177.4 Ω, R_4 = 218.8 Ω, R_5 = 261.2 Ω

- Loop 1 (V_1 = 9.9 V):

VR_1 = 8.96 V
VR_3 = 4.522 V
VR_4 = −5.554 V
Sum of loop voltage ≈ 0.092 V

- Loop 2 (V_2 = 23.97 V):

VR_2 = −8.82 V
VR_3 = −4.521 V
VR_5 = −10.63 V
Sum of loop voltage ≈ 0.01 V

Result

The algebraic sum of voltages in each loop was approximately zero, confirming Kirchhoff's Voltage Law. Currents at junctions can be calculated using KCL if measured, showing that Kirchhoff's Current Law is verified.

Conclusion

Kirchhoff's Voltage Law and Current Law were successfully verified experimentally. Minor deviations are due to instrument tolerances and contact resistance.

Precautions

Ensure all connections are secure.
Use proper ranges on the multimeter to avoid damage.
Avoid short circuits and overheating of resistors.

Sources of Error

Instrument tolerances and calibration
Loose connections
Minor measurement inaccuracies